

DCLab Business Proposal Summary

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Purpose: Provide a short business and technical summary for cofounder agreement.

Executive summary

DCLab is a working environment for data scientists and machine learning engineers. It connects data, experiments, decisions, models, releases, monitoring, costs, and project history so a team can move from a business problem to a reliable model without losing context between tools.

The first product should solve one complete problem well. A customer brings structured business data, defines what it wants to predict, creates and compares models, approves one model, produces batch predictions, and monitors the result. AI helps explain findings and propose work, while DCLab software enforces permissions, budgets, evidence, and approval rules.

Our first commercial test should be a paid design-partner pilot with a small ML team that already has models but struggles with disconnected notebooks, scripts, experiment tools, cloud jobs, documents, and manual production handoffs.

DCLab in one sentence

DCLab helps an ML team move from raw business data to a verified and monitored model while keeping every experiment, decision, cost, and result connected and reviewable.

Founder vision

My vision is for DCLab to become the main working environment where an ML team moves from its first question and dataset to a deployed and monitored result without rebuilding context between tools. Conversation, visual workflow, notebook code cells, experiments, decisions, compute, releases, and monitoring should be connected views of the same project.

I want AI agents to understand project history and help with investigation, planning, coding, experiments, and operations, while DCLab controls identity, permissions, budgets, approvals, evidence, cancellation, and rollback. Over time, a small team should be able to operate with the discipline of a much larger ML organization, deploy on AWS or Google Cloud, and use approved external CPU or GPU capacity without giving up control of project state, secrets, decisions, or audit history.

Customer and business value

The first users are data scientists and ML engineers working with structured data. The likely buyer is a head of data, head of machine learning, chief technology officer, or engineering leader responsible for delivery speed, cost, reliability, and risk.

The strongest early use cases are customer retention, lead scoring, demand prediction, fraud or risk review, and similar recurring batch-prediction problems. DCLab is valuable when a team needs reproducible results, clear decisions, reliable movement into production, and control over compute and AI costs.

The MVP

The MVP is one complete path from data to a monitored batch model.

Product area	First release commitment
Secure project foundation	Workspaces, projects, members, roles, permissions, browser sessions, developer credentials, and audit history
Data access and understanding	CSV and Parquet upload, a small connector set, data profiling, quality checks, leakage checks, and versioned datasets
Experiment workflow	Prediction-goal definition, reviewable experiment plans, controlled training, useful metrics, cost records, and reproducible comparisons
AI assistance	Project-aware explanations, investigations, and proposals with budgets, stop conditions, approved tools, and human control over material changes
Notebook and developer access	Isolated Python notebook sessions plus a versioned API, Python SDK, command line, and controlled integration interface
Model operations	Verified model package, authorized batch predictions, monitoring, investigation, rollback, and evidence linked to the released result
Production readiness	Deployment on either AWS or Google Cloud, with logs, metrics, alerts, backups, recovery, security tests, and cost controls

The MVP does not include unrestricted autonomous changes, real-time global serving, autonomous retraining without review, every possible connector, large foundation-model training, or one active customer database spread across two clouds.

Technical MVP

DCLab separates the permanent **platform** from the temporary **runtime**. The platform stores customer identity, project history, permissions, decisions, current state, and audit evidence. The runtime performs notebook code, data checks, training, evaluation, packaging, and prediction. A failed or replaced compute job must never erase or rewrite the authoritative project record.

Technical part	MVP responsibility
PostgreSQL	Store the authoritative product lifecycle, permissions, versions, jobs, approvals, usage, and audit records
Private object storage	Store datasets, model packages, reports, logs, and prediction files with immutable versions and verified digests
Job system	Queue, limit, cancel, retry, and recover long-running work without creating duplicate results
ML evidence services	Use private MLflow and validation or monitoring tools behind DCLab services while DCLab retains approval and release authority
Notebook runtime	Start with an isolated Jupyter Server session for each authorized user, with idle timeout, absolute expiry, quotas, network policy, cancellation, and cleanup
Cloud deployment	Run the same application and Kubernetes workload contract on either AWS or Google Cloud using short-lived workload identity and no static cloud keys
Optional external compute	Prove home-cloud CPU execution and one external GPU provider first; add Runpod, Railway, Lambda GPU Cloud, Vast.ai, and Nebius only through tested provider adapters

Jupyter Enterprise Gateway is a later scaling option. It should be introduced when shared multi-cluster kernel routing is needed and after its security, cancellation, lifecycle, and recovery behavior passes DCLab's runtime tests.

After the MVP

Later development should follow paid customer demand rather than expand every technical possibility at once.

Area	Direction after validation
Prediction delivery	Low-latency online endpoints, scheduled services, streaming inputs, autoscaling, and stronger traffic controls
Model development	More model families, distributed training, deeper GPU support, advanced tuning, feature reuse, and larger datasets
Runtime scale	Enterprise Gateway or an equivalent remote-kernel layer, larger worker fleets, advanced scheduling, and capacity reservations

Area	Direction after validation
Data ecosystem	More databases, warehouses, SaaS applications, event sources, private connections, and reviewed partner extensions
Automation	Policy-approved retraining, evaluation, release preparation, and rollback recommendations with stronger simulation and approval controls
Enterprise readiness	Single sign-on, automated user provisioning, customer-managed keys, private networking, regional controls, compliance evidence, and support commitments

Business model and market test

The initial business model should combine a platform subscription with usage charges for compute, storage, notebook sessions, batch predictions, and AI models. Enterprise terms can later cover private deployment, stronger identity and security controls, support commitments, and custom integrations.

We should recruit three to five design partners, choose one real structured-data use case for each, measure the current process, run the complete DCLab journey, and compare delivery time, repeat work, reproducibility, manual handoffs, user confidence, cost, and willingness to continue. Pricing should be finalized after these pilots show which outcome customers value and what a normal project costs to operate.

Decisions for the founders

1. **Product definition** Approve DCLab as the connected environment for the ML lifecycle, with AI assistance governed by DCLab.
2. **First customer** Approve structured-data ML teams with recurring batch-prediction needs as the initial segment.
3. **MVP promise** Approve one complete journey from data to a verified and monitored batch model.
4. **Commercial test** Approve founder-led design partnerships and a paid pilot before final pricing.
5. **Technical boundary** Approve one AWS or Google Cloud platform home, Jupyter Server first, and separately certified external compute.
6. **Scope boundary** Keep real-time serving, broad connector coverage, autonomous retraining, and large foundation-model training outside the first release.
7. **Ownership** Assign clear founders to customer discovery and engineering delivery, then review customer and product evidence together each week.

Proposed agreement

If we agree, this becomes DCLab's short business definition; the full proposal and roadmap remain the detailed authorities. Near-term work must serve the MVP, a pilot need, or mandatory safety. The next milestone is one design partner completing a real project and choosing to continue.